

TUSHAR KUMAR

Backend Developer | Node.js | TypeScript | Java | Spring Boot
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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

I build backend services, REST APIs, and production applications using Node.js, TypeScript, Java, and Spring Boot. I've worked on payment and transaction systems, aviation workflow platforms, authentication and RBAC, database-backed applications, reporting systems, and cloud deployments using AWS and Docker.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python

Backend: Node.js, Express.js, Spring Boot, Spring Data JPA, Hibernate, NestJS, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, Redis

Testing: JUnit 5, Jest, H2

Cloud/DevOps: AWS EC2, S3, Docker, GitHub Actions, Linux, Nginx, Git

Core CS: Object-Oriented Programming, DSA, Database Design, Authentication, RBAC, Transactions

PROJECTS

[CreatorStore — E-Commerce Backend API](#)

Java 21, Spring Boot 3, Spring Data JPA, Hibernate, PostgreSQL, JUnit 5, Swagger/OpenAPI

- Built a RESTful e-commerce backend using Spring Boot with layered architecture, constructor-based dependency injection, DTO-based API contracts, and centralized exception handling.
- Implemented transactional checkout with stock validation and atomic inventory updates to prevent inconsistent orders during concurrent purchases.
- Designed PostgreSQL/JPA data models and indexes for product catalog and keyword search operations.
- Added OpenAPI documentation and automated unit/integration tests using JUnit 5 and H2

[Agentic RAG Document Intelligence](#)

Node.js, Express, React, PostgreSQL, pgvector, Gemini API, Docker

- Built a full-stack, multi-tenant RAG system that ingests PDF documents, splits content into overlapping chunks, generates 384-dimensional local embeddings using **all-MiniLM-L6-v2**, and stores vectors in PostgreSQL with pgvector.
- Developed Node.js/Express REST APIs for document upload, querying, listing, and deletion, with session-based isolation to keep user workspaces and document data separated.
- Implemented cosine-similarity retrieval to identify relevant document chunks and integrated Google Gemini for context-grounded question answering with source citations.
- Containerized the frontend, backend, and PostgreSQL services using Docker Compose, enabling reproducible multi-service deployment.

PROFESSIONAL EXPERIENCE

Backend Software Engineer (Contract)

TruesTech IT Solution Pvt. Ltd. — Gurugram, Haryana

Dec 2025 – Present

Tech Stack: Node.js, Express.js, MySQL, MongoDB, Redis, Docker, AWS EC2, PM2, Webhooks

- Built backend services and dashboard functionality for pay-in and payout workflows, processing transaction data between payment gateways, internal services, and banking APIs.
- Integrated payment and banking APIs with secure payload encryption/decryption and request/response transformation for transaction processing.
- Engineered idempotent webhook processing and atomic MySQL transactions with row-level locking (SELECT ... FOR UPDATE), eliminating double-crediting and race conditions during concurrent wallet updates

- Implemented Redis-backed asynchronous worker queues for generating large PDF/CSV/Excel transaction reports, offloading heavy I/O from the main HTTP event loop.
- Deployed Node.js services on AWS EC2 using Docker and PM2, configuring multiple worker processes and Redis-backed workers for concurrent transaction processing and background jobs.

Software Engineer Intern

AirFleet Managers — Gurugram, Haryana

Feb 2025 – Aug 2025

Tech Stack: React, Node.js, Express.js, MongoDB, Redis, Docker, AWS EC2, S3, PM2, Nginx

- Developed frontend and backend functionality for an aviation maintenance workflow platform that centralized aircraft project documentation, maintenance status tracking, collaboration, and reporting across airlines, MROs, third-party providers, and internal users.
- Implemented authentication and role-based access control for employees, administrators, airlines, MROs, and third-party users, separating access to sensitive project and maintenance data.
- Extended MongoDB project models to support multiple aircraft maintenance modules and screens, and implemented live status updates allowing stakeholders to track, review, comment on, and update project progress.
- Built multi-format project reporting with PDF, PowerPoint, and Excel exports, including charts for maintenance/project performance summaries.
- Deployed the application on AWS EC2 with S3-based document storage, PM2 process management, Nginx, MongoDB, and GitHub-based deployment workflows; Configured Nginx as a reverse proxy alongside PM2 on AWS EC2, integrating AWS S3 pre-signed URLs for secure aircraft document storage and retrieval.

Web Developer Intern

Sanrachna Foundation — South West Delhi, Delhi

Jun 2024 – Jul 2024

Technologies: React, Node.js, Express.js, MongoDB, REST APIs, Tailwind CSS, WordPress

- Rebuilt an existing research and publication website with a redesigned React-based interface, improved content categorization, structured research-paper pages, and in-page PDF viewing.
- Improved website performance and content discovery while adding newsletter signup and automated notifications/popups for newly published research papers and events.
- Improved content management and publication workflows for a research portal serving 10,000+ monthly users, contributing to a 30% increase in user engagement.

EDUCATION

SGT University — Gurugram, Haryana

Bachelor of Computer Applications (BCA)

Aug 2022 – Aug 2025